Module-2

1. Explain feature selection compare filter & wrapper methods.

Feature selection:

It is the process of selecting the most relevant & useful features.

In machine learning a data set may contain a large no. of features not all features are useful for Predicting the target variable.

Ex; Suppose we want to predict wheather a student will pass an exam. The dataset may contain;

- Attendance
- Internal marks
- Study hours
- Previous marks
- Student name
- Roll no
- Favorite colour

objectives of feature selection;

1. Reduce the no. of features in the dataset

2. Remove irrelevent & redundant features
3. Improve model Performance
4. Reduce overfitting

Types of feature selection methods.

1. filter method
2. wrapped method
3. Embedded method

Advantage of feature selection:

- Reduces dimensionality
- Improves model efficiency
- Reduces overfitting
- marks models easier to interpret

2. Describe the Decision tree algorithm & its advantages.

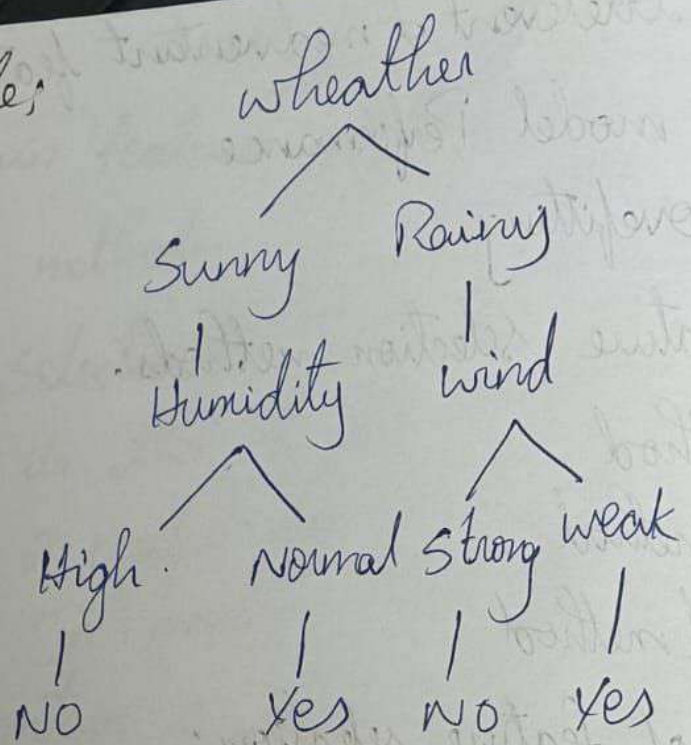
Decision Tree.

A decision tree is a supervised machine learning algorithm used for both classification and regression Problems.

The tree consits of

- Root node
- Branches
- Leaf nodes

Example:



Advantages:
- Easy to understand
- Easy to Interpet
- Handles classification & Regression
- Required less data Preprocessing
- Handles Numerical & categorical Data.

Disadvantages:
- They can easily overfit training data.
- Very deep trees can become complex
- Large trees may be difficult to implement

3. Explain principal component Analysis (PCA) and its applications.

PCA:

It is a dimensionally reduction technique used in machine learning & statistics to reduce the no. of features in a dataset while Presentation as much important as possible

In a dataset many features may contain similar or redundant information. PCA transforms the original features into a new set of variables.

Why PCA needed:

- Increase computational cost
- Make visualization difficult
- Increase the possibility of overfitting
- Make the dataset more complex.

Advantages:

1. Reduce dimensionality
2. Reduces computational complexity

3. Removes Redundancy
4. Helps Redundancy
5. can Reduce Noise.

Applications of PCA:

1. Image Processing
2. Data visualization
3. Machine Learning
4. Face Recognization.